SimpleFT8 — Two Antennas, One Comparison

Resonant antenna, higher band — the strongest diversity evidence in this study — 15m FT8

Period: 2026-04-28 to 2026-05-27 · 15s cycles evaluated: 40342

Summary:

On 15m, ANT1 (Kelemen DP-201510) operates WITHIN its design band — a top-class resonant antenna.
Unlike flat 20m, 15m shows a SIGNIFICANT diversity gain (confidence interval entirely above 0).
15m is the higher resonant band: Faraday rotation scales with f^2 → polarization diversity acts stronger.
When ANT2 (gutter) wins a dual-receive, it does so by +4.3 dB on average.

Note on setup: ANT1 (Kelemen DP-201510 trap dipole) operates within its design band on 15m (20m/15m/10m) — a first-class resonant antenna. ANT2

What do the three modes mean?

Normal (grey): One antenna, no special logic — classic single-antenna setup. Reference baseline.
Diversity Standard (blue): Two antennas. Auto-selects the one with more stations.
Diversity DX (orange): Two antennas. Selects the one with more weak DX signals (below −10 dB).
Rescue (green caps): Stations ANT1 couldn't hear — ANT2 decoded them anyway.

How was it measured?

Setup, time period and a bit of background

The Setup

Measured on 15m FT8 using a FlexRadio FLEX-8400M — two antenna ports, same frequency.
Period: 2026-04-28 to 2026-05-27. Database growing — 15m measured in parallel to 20m/40m.
Cycles: Normal 10895 (5 day/s) | Diversity Standard 11496 (7 day/s) | Diversity DX 17951 (8 day/s).
Note: 15m has a shorter wavelength than 20m → polarization diversity acts even stronger (f^2).

Why not just use the daily average?

Good question — I did it that way at first too. But if one day had only 20 cycles and another had 500, the short day would carry the same weight. That's unfair. So all individual values are pooled and averaged directly — regardless of which day. This gives a picture much closer to reality.

What are 'rescued stations' (Rescue)?

Imagine a station transmitting so weakly that ANT1 receives it below −24 dB.
That is the FT8 decoding threshold — below that, decoding is practically impossible.
ANT2 receives the same station with slightly more signal and decodes it anyway.
We call this 'Rescue'. The green caps in the charts show exactly these stations.
Whether you count them or treat them separately — that's up to you. Both values are in this report.

Where does the raw data come from?

SimpleFT8 writes a small Markdown file per hour containing all cycle values.
No pre-computed averages — raw data only. The analysis script calculates everything fresh.
Want to check? statistics/ in the GitHub repo, all public.

Main Results — Comparison Table

15m FT8 · Pooled mean across all measurement points (database still growing)

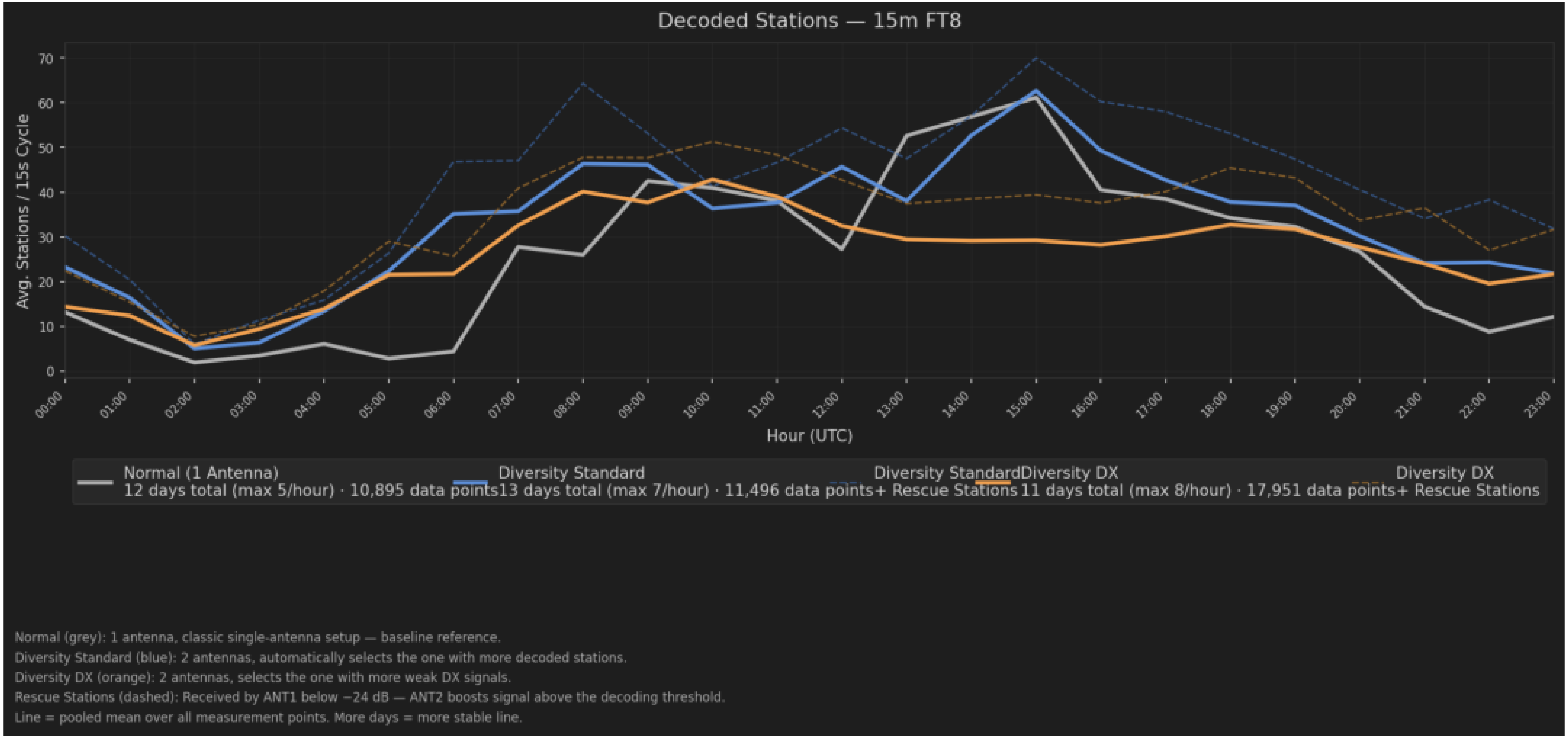
Mode	Avg Sta. /15s Cycle	vs Normal w/o Rescue	vs Normal + Rescue	Rescue only	Meas. Days	Cycles
Normal (1 Antenna	26.8	—	—	—	5	10895
Diversity Standard	35.1	**+31%** **+2–+67%**	**+69%**	**+38%**	7	11496
Diversity DX	28.6	**+7%** **-15–+35%**	**+40%**	**+33%**	8	17951

Avg Sta./15s Cycle: Pooled mean — all measurement points across all days and hours combined. On 15m the database is still growing, so individual UTC hours show more variati

Important: On 15m ANT1 is RESONANT — the diversity gain does NOT come from antenna mismatch. When both antennas receive the same station and ANT2 wins, it does so by

Stations per Hour — all three modes compared

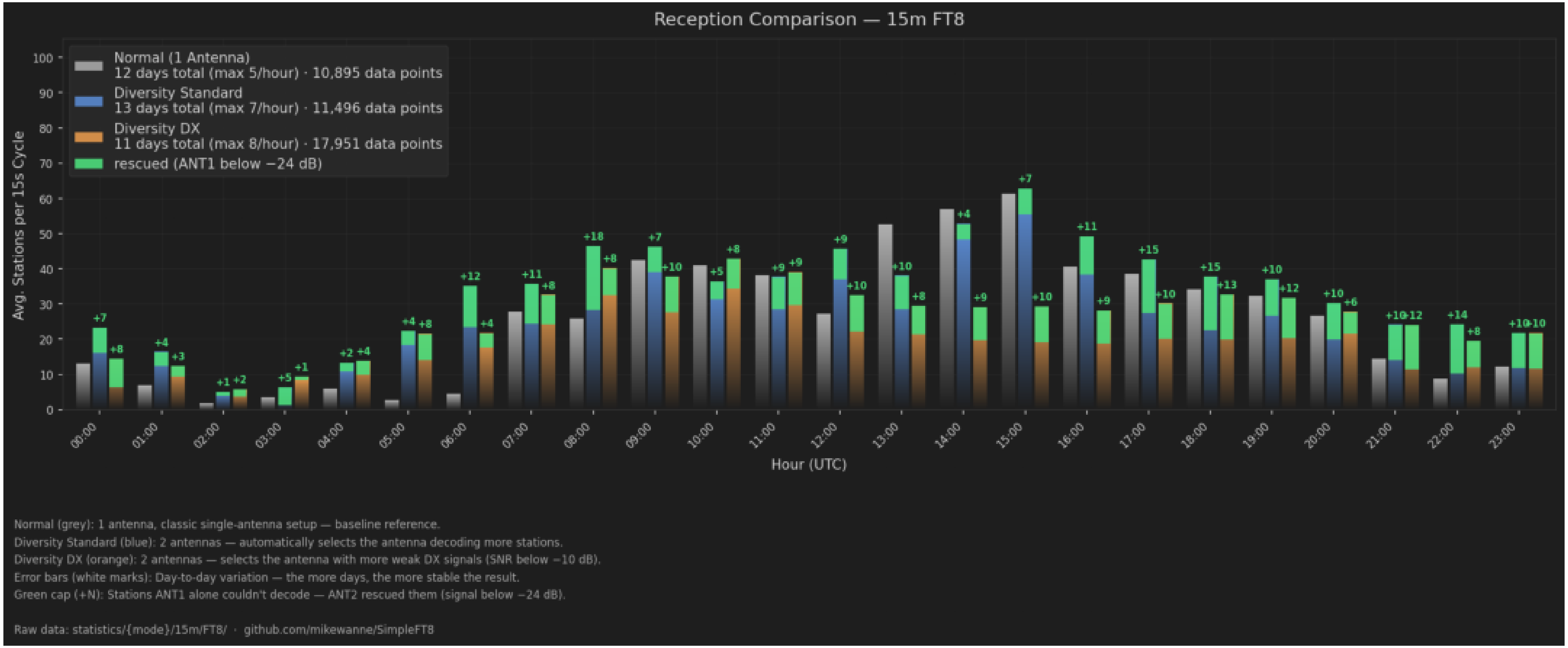
15m FT8 — line = pooled mean over all measurement days. Dashed = + rescue stations.



On 15m diversity stays consistently above Normal — and the gap is clearer than on 20m, even though ANT1 is resonant here too. The shaded band shows variation between measurement days; it narrows with more days.

Direct Comparison — bars per hour, three modes side by side

15m FT8 — Normal (grey) | Diversity Standard (blue) | Diversity DX (orange) | Rescue caps (green)



During daytime peak (12-16 UTC) the diversity advantage shows most clearly. At day→night transition Diversity_DX jumps especially high — DX mode filters for weak signals (SNR<-10dB), and exactly these arrive more often via the quasi-vertical (gutter) at skip-zone edges. White error bars show variation between measurement days.

The Green Caps — count them or not?

Rescue stations: What are they and what do they tell us?

What's this about?

In the diversity chart, small green caps with +N appear above some bars.
These are stations ANT1 couldn't decode — signal below −24 dB.
ANT2 heard and decoded them anyway. Average over the measurement period:
Ø 10.1 stations per hour for Diversity Standard, Ø 9.0/h for Diversity DX.
On 15m the diversity gain comes from a mix: rescue of weak signals AND the polarization/pattern advantage of the second, physically different antenna.

Why should you count them in?

Because the QSO is real for the operator — regardless of whether ANT1 or ANT2 made it possible.
These stations would never have made it into the log with a single antenna.
This is not a measurement artifact — it's exactly the point of running a second antenna.
With Rescue: Standard +69%, DX +40% — that's the full picture.

Why might you leave them out?

SNR values fluctuate within the 15 seconds of a cycle — a station just below −24 dB might be above that threshold in the next cycle.
The −24 dB threshold is an empirical value, not a law of nature.
For a clean comparison with other systems, counting only what ANT1 directly decoded is more conservative.

That's why this report always shows both numbers: without Rescue (the conservative value) and with Rescue (what you get in real operation). Ta

What can you take away from this?

Conclusions and what gets measured next

What is clearly visible:

15m is the most meaningful case in the whole study:
ANT1 is already a top-class resonant antenna here — 'more' was not really to be expected.
Yet ANT2 adds on average +31% to +69%, and the lead is statistically
solid (confidence interval above 0) — unlike flat 20m.
Diversity DX delivers 7% — targeted at weak DX signals.
Higher band, resonant antenna, clear gain: exactly what f^2 theory predicts.
The diversity lead is clearest around 08:00 UTC (+79%) — typically at band-opening and skip-zone edges where polarization diversity acts strong

What cannot be said yet:

The 15m database is still growing — certain UTC hours are underrepresented; the confidence interval narrows with more days.
With a very open band (sunspot maximum) the diversity gain might be even higher — so far only moderate-to-good conditions measured.
The exact dual-receive win rate (like 'X% of cases') is deliberately NOT quoted here — the detail logs only record ANT2 wins, not all comparisons.
Whether this works the same on other transceivers — only tested on the FLEX so far.

What comes next:

More measurement days so per-UTC-hour variation decreases.
Targeted measurements in the still-thin UTC hours — closing the data gaps.
Updates run in parallel to the 20m/40m analysis — all reports maintained together.
This report updates automatically whenever new data comes in.

Raw data: statistics/ in the repo · github.com/mikewanne/SimpleFT8 · DA1MHH